

Tobi Joshua Samuel

Founder & Lead Engineer | Full-Stack, AI & Agentic Systems

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Portfolio | GitHub | Expert Hive Technologies

Professional Summary

Founder, product engineer, and full-stack software developer with 6+ years of experience building production SaaS, education, workforce, real-time, and AI-assisted systems. Founder & Lead Engineer at Expert Hive Technologies, leading architecture, engineering, AI integration, automation, security, and deployment. Strong in Python, Django, React, PostgreSQL, APIs, and cloud systems, with hands-on experience in Cursor/ChatGPT agentic workflows, AI training, data annotation, model evaluation, RLHF-style preference work, scientific coding QA, and machine-learning experimentation.

Professional Experience

Expert Hive Technologies

Oyo State, Nigeria / Remote

Founder & Lead Engineer

2022 – Present

- Lead end-to-end product and engineering strategy across education technology, internal operations, AI-assisted workflows, real-time communication, and customer-facing platforms.
- Architected and built **Expert Hive Tutors**, a tutoring platform supporting student/tutor workflows, bookings, payments, notifications, recommendations, admin operations, and web/mobile experiences.
- Designed **Expert Hive OS**, a school operating system covering administration, attendance, academics, communication, finance, reporting, and analytics in a unified platform.
- Built **Expert Hive Tutor Meet**, a live online-class/video platform built on top of **BigBlueButton (BBB)**, integrating tutoring-session workflows with real-time virtual classrooms.
- Built an internal **recruitment and workforce operations platform** with applicant tracking, scorecards, assessments, interviews, offer workflows, document generation, onboarding, role-based employee workspaces, leave, payroll, reporting, audit trails, and content-task operations.
- Developed AI-assisted product and engineering workflows using **Cursor, ChatGPT, agentic coding, automated testing, code review, prompt-driven implementation, and structured QA** to accelerate delivery while retaining architecture and quality control.
- Design secure REST APIs, authentication/RBAC, payment and notification workflows, PostgreSQL data models, Redis-backed real-time features, and production systems deployed with Railway, AWS, Docker, Cloudflare, CI/CD, and GitHub.

AI Trainer, Data Annotation & Model Evaluation

Project-Based / Remote

- Evaluate AI-generated responses for correctness, reasoning quality, instruction adherence, completeness, formatting, and rubric compliance; provide structured corrections and reviewer feedback.
- Perform **data annotation, labeling, classification, preference ranking, and RLHF-style evaluation** using web-based labeling and review tools across technical and general-domain tasks.
- Perform scientific coding quality control: inspect failed notebook tasks, diagnose evaluator failures, make minimal scientifically correct calibrations, and validate behavior with regression, fresh-namespace, and targeted mutation testing; review prompts, outputs, edge cases, and test behavior to improve evaluation quality.

Sonic Plastics Limited

Edo State, Nigeria (Remote)

Software Developer

2021 – 2022

- Built a Django-based inventory platform for distributed operations and optimized API/data-processing flows, reducing synchronization latency by 45%.
- Implemented AES-256 encryption, role-based access control, and Firestore-backed real-time inventory visibility.
- Mentored junior developers, reviewed code, and refactored application modules to improve maintainability and reduce complexity.

Stag Engineering Limited

Lagos State, Nigeria

Software Developer

2019 – 2021

- Built a React Native task-management application with real-time synchronization and offline support for field workflows.
- Developed and optimized Django REST APIs and PostgreSQL queries, reducing reporting latency by 50%.
- Implemented CI/CD pipelines, automated notifications, and scheduled operational workflows to reduce repetitive manual work.

Selected Products, AI & Engineering Projects

Expert Hive Tutors

Django, DRF, React/React Native, PostgreSQL, Redis

Full-stack tutoring platform with learner/tutor workflows, recommendations, bookings, payments, notifications, support operations, administration, and mobile/web experiences.

Expert Hive OS

Full-Stack SaaS / School Operations

School operating system for administration, attendance, academics, communication, finance, reporting, and analytics, designed as a modular platform for education operations.

Expert Hive Tutor Meet

BigBlueButton, Real-Time Video, Web Integration

Integrated live-class platform built on BigBlueButton for browser-based tutoring sessions and virtual-classroom workflows.

Expert Hive Recruitment & Workforce Platform

ATS, RBAC, Automation, Audit

Internal operations system covering careers, applicant scoring, assessments, interview scheduling/scoring, offer packages, hiring handoff, employee onboarding, workspaces, payroll, leave, reports, documents, and content workflows.

AI-Assisted Customer Support Platform

AI, Workflow Automation, Support Operations

Designed AI-assisted support workflows for customer enquiries, triage, knowledge handling, escalation, and operational follow-up.

Development of a Lightweight AI-Assisted Triage Pipeline for Chest Imaging

Python, Deep Learning, Explainability, Uncertainty & Calibration

Research/preprint on reliability-aware chest X-ray and CT triage using compact models, Grad-CAM explainability, uncertainty estimation, calibration, and low-latency deployment. DOI: 10.13140/RG.2.2.19454.24647.

Real-Time Intrusion Detection System

Python, Machine Learning

Developed a hybrid intrusion-detection workflow combining supervised and unsupervised models with monitoring and automated alerting.

Expert Hive Mobile App

React Native, Firebase, Django

Built a cross-platform tutoring/mobile experience with real-time messaging, push notifications, offline-aware workflows, and backend integration.

React TypeFlare

React, JavaScript, NPM

Published a lightweight reusable typewriter-animation component with reverse-typing effects for React applications.

Vyzio Classifieds & Ads Marketplace

Django REST, React, Payments

Implemented listing lifecycle, moderation workflows, APIs, and secure payment-processing integrations for a marketplace platform.

Technical Skills

AI, Agentic & Model Evaluation: Cursor, ChatGPT, agentic development workflows, prompt engineering, AI-assisted coding, model-response evaluation, RLHF/preference ranking, data annotation, labeling, rubric-based QA, scientific coding evaluation

Languages: Python, JavaScript, TypeScript, R, MATLAB, C#

Backend & APIs: Django, Django REST Framework, Node.js, REST APIs, WebSockets, authentication, RBAC, payments, automation

Frontend & Mobile: React, React Native, HTML5, CSS3, responsive UI development

Data, ML & Analytics: Scikit-learn, TensorFlow, Pandas, NumPy, BigQuery, Tableau, ML evaluation, explainability, uncertainty/calibration concepts

Databases & Real-Time: PostgreSQL, Redis, Firestore, Redis pub/sub, Socket.io, BigBlueButton

Cloud & DevOps: Railway, AWS (EC2, S3), Docker, Cloudflare, GitHub, CI/CD, production deployment and debugging

Engineering Practice: System design, API architecture, testing, TDD, code review, Git, documentation, LaTeX/TikZ, technical research

Education

University of Lagos

Lagos, Nigeria

B.Sc. Mechanical Engineering

November 2021

Leadership & Professional Strengths

Product strategy, technical leadership, founder-led engineering, mentoring, cross-functional delivery, customer-focused product development, requirements translation, rapid prototyping, production troubleshooting, and continuous improvement.